PART A — QUESTIONS

AF16-001
Question:
Solve: $18x + 9 = 117$
A) 5
B) 6
C) 7
D) 8

AF16-002
Question:
If $16n - 8 = 104$, what is n?
A) 5
B) 6
C) 7
D) 8

AF16-003
Question:
Simplify: $9(2x - 4) + 6x$
A) $24x - 36$
B) $18x - 36$
C) $24x - 4$
D) $15x - 36$

AF16-004
Question:
A company charges $27 per service call plus a $40 account fee. Which expression represents the total cost for c service calls?
A) $27c + 40$
B) $40c + 27$
C) $67c$
D) $40 - 27c$

AF16-005
Question:
If $f(x) = 13x - 25$, what is $f(8)$?
A) 79
B) 89
C) 104
D) 129

AF16-006
Question:
Which ordered pair satisfies the equation $y = 11x + 3$?
A) (1, 13)
B) (2, 25)
C) (3, 35)
D) (4, 46)

AF16-007
Question:
Solve: 17(x - 5) = 102
A) 9
B) 10
C) 11
D) 12

AF16-008
Question:
A pattern begins: 9, 24, 39, 54, ... What is the next number?
A) 66
B) 68
C) 69
D) 70

AF16-009
Question:
Solve: x/30 = 3
A) 33
B) 60
C) 90
D) 120

AF16-010
Question:
A line has a slope of -13 and crosses the y-axis at 10. Which equation represents the line?
A) y = 10x - 13
B) y = -13x + 10
C) y = 13x + 10
D) y = -10x + 13

AF16-011
Question:
If 7 crates contain 189 parts, how many parts are in 5 crates at the same rate?
A) 120
B) 125
C) 135
D) 145

AF16-012
Question:
Solve: 150 - 21x = 66
A) 3
B) 4
C) 5
D) 6

AF16-013
Question:
Which expression is equivalent to 63m - 28m - 15?

A) 35m - 15
B) 91m - 15
C) 35m + 15
D) 63m - 43

AF16-014
Question:
A shelf starts with 875 parts. Each job uses 25 parts. Which equation gives the number of parts P left after j jobs?
A) P = 875j - 25
B) P = 25j + 875
C) P = 875 - 25j
D) P = 25j - 875

AF16-015
Question:
If y = -15x + 170, what is y when x = 9?
A) 25
B) 35
C) 135
D) 305

AF16-016
Question:
Solve: 26x - 28 = 17x + 62
A) 8
B) 9
C) 10
D) 11

AF16-017
Question:
Which value of x makes 14x + 6 greater than 132?
A) 8
B) 9
C) 10
D) 7

AF16-018
Question:
A table shows a linear relationship.

x: 0, 1, 2, 3
y: 4, 20, 36, 52

What is the rule?
A) y = 4x + 16
B) y = 16x + 4
C) y = 20x - 16
D) y = x + 4

AF16-019
Question:
Solve: 13(x + 4) = 10x + 82
A) 8
B) 9
C) 10
D) 11

AF16-020
Question:
If a = -12 and b = 9, what is 3a + 5b?
A) -81
B) -9
C) 9
D) 81

AF16-021
Question:
Which point is located in Quadrant II?
A) (12, 6)
B) (-12, 6)
C) (-12, -6)
D) (12, -6)

AF16-022
Question:
A worker packs 168 items in 12 minutes. At the same rate, how many items can the worker pack in 9 minutes?
A) 112
B) 120
C) 126
D) 132

AF16-023
Question:
Solve for q: T = 22q + 44
A) q = T - 44
B) q = (T - 44)/22
C) q = 22(T - 44)
D) q = T/22 + 44

AF16-024
Question:
What is the slope of a line that goes through (4, 6) and (10, 60)?
A) 7
B) 8
C) 9
D) 54

AF16-025
Question:

Simplify: $45x + 11 - 34x - 26$
A) $11x - 15$
B) $79x - 15$
C) $11x + 37$
D) $79x + 37$

AF16-026
Question:
If $g(x) = x^2 - 9x$, what is $g(13)$?
A) 39
B) 52
C) 117
D) 169

AF16-027
Question:
A sequence follows the rule “multiply by 3, then subtract 11.” If the first term is 12, what is the second term?
A) 25
B) 27
C) 36
D) 47

AF16-028
Question:
Solve: $16(x - 4) + 10 = 106$
A) 8
B) 9
C) 10
D) 11

AF16-029
Question:
A line crosses the y-axis at -15 and rises 7 units for every 1 unit to the right. Which equation represents the line?
A) $y = -15x + 7$
B) $y = 7x - 15$
C) $y = -7x - 15$
D) $y = 15x + 7$

AF16-030
Question:
If $20/25 = x/75$, what is x?
A) 50
B) 55
C) 60
D) 65

AF16-031
Question:
Solve: $-20x + 60 = -140$

A) -10
B) -6
C) 6
D) 10

AF16-032
Question:
A warehouse starts with 2,400 clips. Each package uses 120 clips. Which expression gives the number of clips left after p packages?
A) 2,400 + 120p
B) 2,400 - 120p
C) 120p - 2,400
D) 2,400 ÷ 120p

AF16-033
Question:
Which equation has x = 21 as its solution?
A) 2x + 15 = 55
B) 4x - 18 = 66
C) x/7 + 8 = 12
D) 3x - 10 = 50

PART B — ANSWER KEY + EXPLANATIONS

AF16-001 — Correct: B — Explanation:
Solve 18x + 9 = 117. Subtract 9 from both sides: 18x = 108. Divide by 18: x = 6. The most common mistake is subtracting 9 correctly but forgetting to divide by 18.

AF16-002 — Correct: C — Explanation:
Start with 16n - 8 = 104. Add 8 to both sides: 16n = 112. Divide by 16: n = 7. A common mistake is subtracting 8 instead of adding it.

AF16-003 — Correct: A — Explanation:
Distribute first: 9(2x - 4) = 18x - 36. Then add 6x: 18x + 6x - 36 = 24x - 36. A common mistake is forgetting to multiply -4 by 9.

AF16-004 — Correct: A — Explanation:
The company charges $27 for each service call, so that part is 27c. The $40 account fee is added one time. The total cost is 27c + 40. A common mistake is treating the account fee as the per-call charge.

AF16-005 — Correct: A — Explanation:
Substitute 8 for x: f(8) = 13(8) - 25 = 104 - 25 = 79. A common mistake is adding 25 instead of subtracting it.

AF16-006 — Correct: B — Explanation:
Test the choices in y = 11x + 3. For (2, 25), 11(2) + 3 = 22 + 3 = 25, which matches the y-value. A common mistake is choosing a point that is close but does not make the equation true.

AF16-007 — Correct: C — Explanation:
Solve $17(x - 5) = 102$. Divide both sides by 17: $x - 5 = 6$. Add 5: $x = 11$. A common mistake is adding 5 before undoing the multiplication by 17.

AF16-008 — Correct: C — Explanation:
The pattern increases by 15 each time: 9, 24, 39, 54. Add 15 to 54: 69. A common mistake is assuming the increase is 14 because the numbers are close together.

AF16-009 — Correct: C — Explanation:
$x/30 = 3$ means x divided by 30 equals 3. Multiply both sides by 30: $x = 90$. A common mistake is adding 30 and 3 instead of multiplying.

AF16-010 — Correct: B — Explanation:
Slope-intercept form is $y = mx + b$. The slope is -13 and the y-intercept is 10, so the equation is $y = -13x + 10$. A common mistake is switching the slope and y-intercept.

AF16-011 — Correct: C — Explanation:
Find the unit rate: 189 parts ÷ 7 crates = 27 parts per crate. For 5 crates: $27 \times 5 = 135$. A common mistake is multiplying 189 by 5 without finding the rate first.

AF16-012 — Correct: B — Explanation:
Solve $150 - 21x = 66$. Subtract 150 from both sides: $-21x = -84$. Divide by -21: $x = 4$. A common mistake is losing the negative sign after subtracting 150.

AF16-013 — Correct: A — Explanation:
Combine like terms: $63m - 28m = 35m$. The constant -15 stays the same. The expression becomes $35m - 15$. A common mistake is combining the constant with the variable terms.

AF16-014 — Correct: C — Explanation:
The shelf starts with 875 parts. Each job uses 25 parts, so j jobs use 25j parts. The number left is $P = 875 - 25j$. A common mistake is adding 25j even though parts are being used.

AF16-015 — Correct: B — Explanation:
Substitute $x = 9$ into $y = -15x + 170$: $y = -15(9) + 170 = -135 + 170 = 35$. A common mistake is treating -15(9) as positive 135.

AF16-016 — Correct: C — Explanation:
Solve $26x - 28 = 17x + 62$. Subtract 17x: $9x - 28 = 62$. Add 28: $9x = 90$. Divide by 9: $x = 10$. A common mistake is adding 28 to only one side.

AF16-017 — Correct: C — Explanation:
Solve $14x + 6 > 132$. Subtract 6: $14x > 126$. Divide by 14: $x > 9$. The only listed value greater than 9 is 10. A common mistake is choosing 9, but $14(9) + 6 = 132$, not greater than 132.

AF16-018 — Correct: B — Explanation:
The y-values increase by 16 each time x increases by 1, so the slope is 16. When $x = 0$, $y = 4$, so the y-intercept is 4. The rule is $y = 16x + 4$. A common mistake is using 4 as the slope because it is the first y-value.

AF16-019 — Correct: C — Explanation:
Distribute: $13x + 52 = 10x + 82$. Subtract 10x: $3x + 52 = 82$. Subtract 52: $3x = 30$. Divide by 3: $x = 10$. A common mistake is forgetting to multiply 4 by 13.

AF16-020 — Correct: C — Explanation:
Substitute $a = -12$ and $b = 9$: $3a + 5b = 3(-12) + 5(9) = -36 + 45 = 9$. A common mistake is treating 3(-12) as positive 36.

AF16-021 — Correct: B — Explanation:
Quadrant II has a negative x-coordinate and a positive y-coordinate. The point (-12, 6) fits this description. A common mistake is choosing (12, -6), which is Quadrant IV.

AF16-022 — Correct: C — Explanation:
168 items in 12 minutes means $168 \div 12 = 14$ items per minute. In 9 minutes: $14 \times 9 = 126$. A common mistake is multiplying 168 by 9 without finding the rate first.

AF16-023 — Correct: B — Explanation:
Start with $T = 22q + 44$. Subtract 44 from both sides: $T - 44 = 22q$. Divide by 22: $q = (T - 44)/22$. A common mistake is dividing T by 22 first and then subtracting 44.

AF16-024 — Correct: C — Explanation:
Slope equals change in y divided by change in x. From (4, 6) to (10, 60), y increases by 54 and x increases by 6. The slope is $54 \div 6 = 9$. A common mistake is using 54 as the slope without dividing by the change in x.

AF16-025 — Correct: A — Explanation:
Combine like terms: $45x - 34x = 11x$, and $11 - 26 = -15$. The simplified expression is $11x - 15$. A common mistake is writing $11x + 15$ because the negative sign before 26 is missed.

AF16-026 — Correct: B — Explanation:
Substitute 13 for x: $g(13) = 13^2 - 9(13) = 169 - 117 = 52$. A common mistake is calculating 13^2 as 26 instead of 169.

AF16-027 — Correct: A — Explanation:
Start with 12. Multiply by 3: 36. Then subtract 11: 25. A common mistake is subtracting 11 first and then multiplying.

AF16-028 — Correct: C — Explanation:
Solve $16(x - 4) + 10 = 106$. Distribute: $16x - 64 + 10 = 106$. Combine: $16x - 54 = 106$. Add 54: $16x = 160$. Divide by 16: $x = 10$. A common mistake is forgetting to combine -64 and +10.

AF16-029 — Correct: B — Explanation:
The line rises 7 units for every 1 unit to the right, so the slope is 7. It crosses the y-axis at -15, so the equation is $y = 7x - 15$. A common mistake is switching the slope and intercept.

AF16-030 — Correct: C — Explanation:
Use the proportion $20/25 = x/75$. Since $25 \times 3 = 75$, multiply 20 by 3: $x = 60$. A common mistake is adding 50 to the numerator because 25 becomes 75.

AF16-031 — Correct: D — Explanation:
Solve $-20x + 60 = -140$. Subtract 60 from both sides: $-20x = -200$. Divide by -20: $x = 10$. A common mistake is giving -10 because both sides contain negative numbers.

AF16-032 — Correct: B — Explanation:
The warehouse starts with 2,400 clips. Each package uses 120 clips, so p packages use 120p clips. The number left is 2,400 - 120p. A common mistake is adding 120p even though clips are being used.

AF16-033 — Correct: B — Explanation:
Test $x = 21$. In choice B, $4(21) - 18 = 84 - 18 = 66$, so the equation is true. A common mistake is choosing a nearby equation without checking both sides exactly.